

Project Proposal

3D Ship Battle Simulation Using OpenGL

Name: Tawhidul Hasan
Roll: 2107004
Course: Image Processing and Computer Graphics Laboratory

Introduction

This project will create a small **3D naval battle scene** inspired by *Assassin's Creed IV: Black Flag*, *Pirates of the Caribbean*, and historical naval warfare. The main purpose is not to create a complete game, but to clearly demonstrate important **Computer Graphics concepts using OpenGL**.

The scene will contain two or more ships moving on a sea surface and attacking each other using cannons. The entire simulation will run continuously inside the OpenGL render loop. All movements and animations will be calculated during program execution, so no pre-computed animation will be used.

Project Description

The scene will show ships moving and rotating on the sea while fighting each other. A ship will be able to rotate its cannon and fire a cannonball toward another ship. The cannonball will visibly travel through the 3D scene so that its movement can clearly be understood.

Simple animations such as ship rocking, cannon rotation, cannon recoil, and cannonball movement may also be included. These animations will demonstrate continuous motion and transformation in a 3D environment.

Computer Graphics Concepts

The project will demonstrate the following concepts:

- **Translation** – movement of ships and cannonballs.
- **Rotation** – ship direction, cannon aiming, and ship rocking.
- **Scaling** – construction and resizing of ship components.
- **Transformation Composition** – combining translation, rotation, and scaling.
- **Continuous Animation** – all motion will be updated inside the render loop.

Lighting and Shading

Proper lighting will be maintained throughout the scene. A directional light source will represent sunlight or moonlight. The project will use the **Phong lighting model**, including ambient, diffuse, and specular components.

Specular reflection will be used to create visible highlights on surfaces such as the ships and sea. This will clearly demonstrate lighting and shading concepts instead of using lighting only for visual decoration.

Scene and Main Objects

To avoid unnecessary calculations and excessive rendering, the scene will remain simple. The main objects will include:

- Sea surface
- Two or more 3D ships
- Ship cannons
- Cannonballs
- Simple sky/background
- Light source

The ships will be constructed using simple 3D shapes and OpenGL transformations instead of unnecessarily complex models.

Tools and Implementation

The project will be developed using **C/C++ and OpenGL**. OpenGL will be used for 3D rendering, transformations, camera control, animation, lighting, Phong shading, and specular reflection.

Keyboard controls may also be added for moving a ship, changing its direction, rotating the cannon, and firing cannonballs.

Expected Outcome

The final project will be a continuously running 3D ship battle simulation where ships move and rotate, cannons aim and fire, cannonballs travel through the scene, and objects respond properly to lighting.

The project will clearly demonstrate **3D rendering, translation, rotation, scaling, transformation composition, continuous animation, Phong lighting, and specular reflection using OpenGL**.